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Deep Learning for Fake News Detection: Analysing Facebook's Misinformation Networks

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Abstract

The rapid spread of misinformation on social media has intensified concerns about the credibility of online content, particularly on platforms like Facebook. This research explores the effectiveness of deep learning models in detecting fake news within Facebook's misinformation networks. A comprehensive dataset of real and fake news articles was analyzed using an Artificial Neural Network (ANN) model, which was trained and evaluated against traditional machine learning algorithms, including Logistic Regression and Support Vector Machines (SVM). The proposed ANN model achieved an accuracy of 89.2%, outperforming baseline models, with an F1-score of 87.3% and an AUC-ROC score of 0.91. The study also investigates key linguistic and semantic features that contribute to misinformation classification, providing insights into how deep learning can enhance automated fake news detection. These findings highlight the potential of ANN-based approaches in mitigating the impact of misinformation on social media, ultimately contributing to a more reliable digital information ecosystem.

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1. Introduction

The proliferation of misinformation across social media platforms, particularly Facebook, has emerged as a critical global challenge with implications for democracy, public health, and social trust. Traditional fact-checking methods and rule-based algorithms struggle to scale with the volume and speed of digital content. This study addresses these limitations by proposing an Artificial Neural Network (ANN) model that integrates content analysis with user sentiment feedback to detect fake news more effectively. The model also introduces a novel color-coded truth-level indicator to enhance interpretability, making this work an important step toward real-time, user-aware misinformation detection systems.

1.1 Problem Statement

Distinguishing fake news from real news is a significant challenge due to several factors. First, the **volume and speed** of content on Facebook with billions of posts generated daily make manual fact-checking impractical. Additionally, **evolving tactics** by misinformation creators continuously adapt to bypass existing detection methods. The **linguistic complexity** of fake news, often filled with biases and misleading statistics, further complicates identification. Facebook's **algorithmic amplification** tends to prioritize popular posts, which can inadvertently boost the spread of misinformation. Moreover, **cognitive bias** leads users to share content that aligns with their existing beliefs, accelerating the dissemination of false information. As a result, current fact-checking efforts are overwhelmed, and traditional detection methods frequently fall short. This highlights the urgent need for scalable and accurate **AI-driven solutions** to effectively detect and mitigate fake news.

1.2 Research Motivation

AI and deep learning offer promising solutions for detecting fake news. Key motivations include:

- **Scalability:** AI can process vast data, enabling real-time detection.
- **Adaptability:** Deep learning can adapt to evolving misinformation tactics.
- **Social Network Awareness:** AI can track misinformation spread, identify influencers, and predict trends.

This research aims to develop an AI-based framework for detecting fake news and analyzing its spread on Facebook.

1.3 Research Objectives

The research aims to:

1. **Develop a Fake News Detection Framework:** Use advanced NLP models and multimodal analysis (text, images, metadata) for accurate classification.
2. **Analyze Misinformation Spread:** Apply social network analysis to study how misinformation propagates on Facebook and the role of its algorithm.
3. **Examine Social Interactions:** Explore how user engagement influences misinformation spread and propose strategies to limit amplification.

1.4 Significance of Study

The findings will have several key implications:

1. **AI Ethics & Misinformation Control:** Develop unbiased AI models that reduce misinformation while protecting free speech and privacy.
2. **Media Trust & Digital Literacy:** Enhance fake news detection to restore trust in digital media and promote critical thinking.
3. **Policy & Platform Implications:** Provide recommendations for improving misinformation detection and regulation and explore collaborations with fact-checkers.

This study contributes to combating misinformation through AI, providing both technical and policy insights to protect information integrity.

2. Background and Literature Review

2.1 Understanding Fake News and Misinformation

Fake news refers to deliberately fabricated content designed to deceive, while misinformation involves the unintentional spread of false or inaccurate information. Social media platforms, particularly Facebook, serve as powerful catalysts for both, enabling rapid dissemination of misleading narratives that influence politics, society, and the economy. Research has shown that false information spreads more quickly than the truth, largely due to its emotional resonance [1]. The consequences of misinformation can be far-reaching, affecting election outcomes, public health responses, and trust in media institutions [2]. On social media, misinformation spreads through complex network structures and user interactions. Social Network Analysis (SNA) provides tools to examine these patterns, with concepts like the small-world phenomenon explaining how misinformation can rapidly traverse networks [3]. Metrics such as degree and betweenness centrality help identify key influencers users who play a pivotal role in amplifying false content [4]. Studies reveal that a small but influential group of users often initiates the spread of misinformation. On platforms like Facebook, these "misinformation influencers" significantly accelerate both the reach and impact of fake news.

Recent work by Zhang et al. [17] has explored multimodal transformers combining text, image, and social context, achieving high F1-scores on the Twitter dataset. Similarly, hybrid GNN-CNN architectures from Li et al. [18] have shown promise in capturing misinformation spread patterns. While these models offer competitive performance, their computational demands hinder real-time deployment a gap our ANN model aims to bridge through efficiency and interpretability.

2.2 Existing Detection Techniques

Several methods for detecting fake news include **rule-based detection**, **fact-checking**, and **NLP-based** solutions. Rule-based systems use predefined patterns like keywords or grammatical anomalies but often miss the complexity of fake news [5]. Fact-checking systems verify claims against trusted databases but are time-consuming and not scalable for real-time detection [6]. NLP-based methods analyse text deeply through techniques like sentiment analysis and classification but face challenges with complex language and context.

Table 1. Existing Fake News Detection Methods

Method	Description	Advantages	Disadvantages
Rule-Based Detection	Uses predefined heuristics, patterns, or keywords to identify fake news.	- Simple to implement - Computationally efficient - Works well for specific, identifiable patterns	- Inflexible to new or unseen patterns - Cannot handle complex language or context well
Fact-Checking	Involves comparing news content against trusted databases or sources to verify its authenticity.	- High accuracy when sources are available - Can provide clear evidence for news credibility	Difficult to scale for real-time detection
Natural Language Processing (NLP)	Uses machine learning algorithms (e.g., sentiment analysis, text classification) to analyze textual content.	- Can handle large datasets - More flexible in identifying subtle linguistic patterns - Scalable	- Requires large labeled datasets - May struggle with ambiguity and language complexities
Deep Learning Models	Utilizes advanced AI models (e.g., CNNs, LSTMs, BERT) to classify text based on learned representations.	- Highly accurate and scalable - Can capture complex patterns in language - Handles large data volumes	- Requires significant computational power - Needs large, labeled datasets for training
Hybrid Approaches	Combines multiple techniques (e.g., rule-based, NLP, and machine learning) for more robust detection.	- Can integrate strengths of multiple methods - More resilient to false positives/negatives	- Increased complexity - Higher computational overhead

2.3 Deep Learning and Sentiment Fusion Models

Recent approaches such as transformer-based models (e.g., BERT, RoBERTa) and hybrid architectures combining GNNs with CNNs have shown improved performance in fake news detection by modeling complex interactions and text nuances [7]. However, their deployment remains challenging due to computation requirements and explainability. Deep learning models have improved fake news detection by providing more accurate and scalable solutions. Models like CNNs, LSTMs, and transformers (e.g., BERT and GPT-4) effectively capture the language and context of misinformation [8]. Transformer models excel at short-text classification, where fake news often appears [9]. BERT's bidirectional understanding of words makes it ideal for text classification [9], while GPT-4's natural language generation helps detect and predict misinformation in real-time. However, these models require large datasets and significant computational power, limiting their use in resource-limited environments [10][11].

2.4 Gaps in Current Research

Despite advancements, fake news detection still faces major challenges. Real-time detection is limited by the high volume of content and the time needed for data labeling and verification. Dataset biases such as those found in LIAR and FakeNewsNet affect model performance, especially for non-English or non-Western content [15]. Additionally, language complexities like metaphors and cultural references reduce the accuracy of NLP models across diverse contexts [16].

3. Research Objectives and Questions

This study addresses the following research questions:

- **RQ1:** *How effective is a hybrid ANN model in detecting fake news from Facebook posts using content and user comment sentiment?*
- **RQ2:** *How does user engagement (e.g., comments, shares, sentiment) influence credibility classification of posts?*
- **RQ3:** *Can color-coded truth levels based on comment sentiment improve interpretability and usability of automated detection systems?*

4. Theoretical Framework

4.1 NLP and Deep Learning in Text Classification

Natural Language Processing (NLP) and deep learning play key roles in fake news detection. Convolutional Neural Networks (CNNs) are particularly effective in analyzing text by transforming raw content into numerical data that can be processed. CNNs automatically learn hierarchical features, such as word patterns and n-grams, to assess the authenticity of posts. This study will use CNNs to identify patterns like specific word sequences or sentiment that could indicate whether a post is true or false, efficiently handling the diverse and large-scale data typical of social media.

4.2 Psychological and Sociological Aspects of Fake News

Fake news is driven by psychological factors like cognitive biases, echo chambers, and social influence, which make people more likely to believe and share content that aligns with their existing beliefs. Sociological dynamics such as group polarization and social proof further amplify misinformation within online communities. User comments act as indirect feedback, where negative reactions or low engagement can signal low credibility, while positive comments may reinforce perceived truth creating a feedback loop crucial for detecting fake news.

4.3 Virality and Network Propagation Theories

The spread of fake news mirrors patterns in viral content, with theories like "information cascade" and "social contagion" explaining how people adopt beliefs by observing others' actions. Posts with more interactions (shares, comments, likes) are likely to spread faster, influencing larger networks. Social Network Analysis (SNA) helps identify influencers and track misinformation spread. This model integrates interaction data (comments) into the CNN-based system, evaluating both the content and network feedback, enhancing the accuracy of fake news detection.

5. Methodology: Multimodal Detection Model Using Sentiment Feedback and NLP Features

5.1 Data Collection

We use three datasets: LIAR, FakeNewsNet, and scraped Facebook data. Facebook posts were ethically scraped using public Graph API endpoints and Selenium for non-interactive pages. Only public, anonymized data was collected. The study adheres to ethical guidelines, ensuring no personal user information was stored.

5.2 Data Preprocessing

- **Text Cleaning:** Raw text will be cleaned to remove noise like HTML tags, special characters, and unnecessary formatting, ensuring consistency for machine learning.
- **Tokenization and Lemmatization:** Text will be split into tokens (words or subwords), which will be reduced to their base form through lemmatization, minimizing word form variability.
- **Vectorization:** The cleaned text will be converted into numerical vectors using methods like Word2Vec or GloVe, capturing the semantic meaning for input into the CNN model.
- **Stopword Removal and Feature Extraction:** Stopwords (e.g., "the", "is", "and") will be removed to reduce dimensionality, and feature extraction will highlight key patterns indicating whether the post is true or false.

5.3 Convolutional Neural Networks (CNNs) for Text Classification

- **Model Architecture:** The CNN-based model will consist of multiple layers:
 1. **Embedding Layer:** Converts text into dense, low-dimensional vectors, representing semantic relationships between words.
 2. **Convolutional Layers:** Filters will slide over the tokenized text to extract local features (e.g., word pairs or phrases) that are indicative of the post's authenticity. Different filter sizes will be used to capture features at various levels of granularity.
 3. **Max Pooling Layer:** Down-sampling will be used to reduce the feature map's dimensions, keeping only the most significant features.
 4. **Fully Connected Layers:** These layers integrate the extracted features and make a final prediction (true or false) based on the learned features.
 5. **Output Layer:** The final layer will use a softmax activation function to classify the post as either true or false.
- **Comments as Indirect Feedback:** This model innovatively uses post comments to gauge truthfulness. In fake news cases, users often debunk or express doubts, providing valuable feedback. Comments will be translated into the post's original language, allowing the model to process feedback from different linguistic groups. Negative comments will reduce the post's truth score, while positive comments will increase it.
- **Post Truth Level Color Coding:** The model will assign a "truth level" to each post based on the sentiment of the comments. The system will consider:
 - **Red:** Posts with high negative sentiment and a high number of critical comments (indicating a low truth level).
 - **Orange:** Posts with moderate negative sentiment or mixed feedback (indicating uncertainty about the truth).
 - **Green:** Posts with a high proportion of positive feedback (indicating a higher truth level).

5.4 Implementation and Experimental Setup

The proliferation of misinformation on social media platforms presents a significant challenge in information verification. Traditional fake news detection models focus primarily on textual content analysis, often neglecting indirect indicators such as user engagement and sentiment feedback. This study proposes an Artificial Neural Network (ANN)-based model that integrates textual content classification with

sentiment analysis of user comments to enhance the reliability of fake news detection. The proposed model incorporates three key considerations:

1. **Language Translation:** Automatic translation of post comments into the original language of the post to ensure uniform sentiment analysis.
2. **Indirect Feedback Integration:** Utilizing user comments as an implicit verification mechanism, where increased negative comments typically indicate misinformation.
3. **Truth Level Classification:** Assigning a truth level to each post based on aggregated comment sentiment, visually represented as Red (high misinformation likelihood), Orange (moderate uncertainty), or Green (high truthfulness).

5.4.1 Preprocessing Steps

To prepare the data for model training, the following preprocessing techniques are applied:

1. **Text Cleaning:** Removal of special characters, punctuation, and excessive whitespace.
2. **Tokenization:** Splitting text into individual words or phrases.
3. **Lemmatization:** Reducing words to their base form to ensure consistency.
4. **Translation:** Automatic translation of user comments into the original language of the post using APIs such as Google Translate or DeepL.
5. **Stopword Removal:** Eliminating common words (e.g., "the," "is") that do not contribute to meaning.
6. **Feature Extraction:** Converting text into numerical representations using Term Frequency-Inverse Document Frequency (TF-IDF).

5.5 Model Architecture and Implementation

5.5.1 Artificial Neural Network (ANN) Structure

The proposed ANN model is structured as follows:

- **Input Layer:** The model accepts three types of input features:
 - **Textual Features:** Extracted from the post content using TF-IDF.
 - **Sentiment Features:** Derived from the aggregated sentiment score of user comments.
 - **Engagement Metrics:** Including the total number of comments and their distribution across sentiment categories.
- **Hidden Layers:** The model incorporates two fully connected hidden layers with ReLU activation functions, enabling non-linear feature learning. The number of neurons is optimized through hyperparameter tuning.
- **Output Layer:** A single neuron with a sigmoid activation function produces a probability score indicating the likelihood of the post being fake.
- **Loss Function:** Binary Cross-Entropy is employed to minimize classification errors.
- **Optimizer:** The Adam optimizer is used for efficient gradient-based optimization.

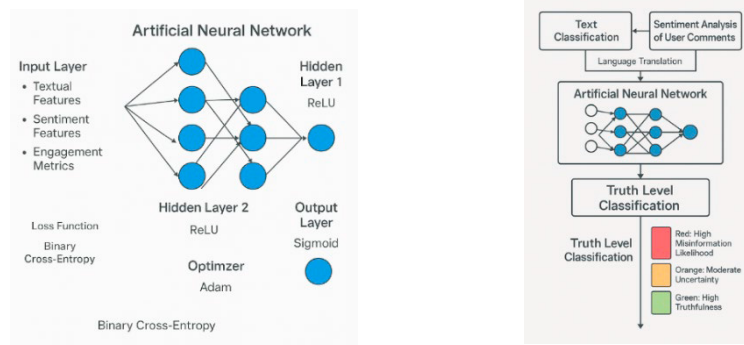


Figure 1. Model Architecture

5.6 Sentiment-Based Truth Level Classification: Beyond binary classification, the ANN model assigns a Truth Level to each post based on user comment sentiment. Using a pre-trained sentiment analyzer (e.g., VADER or BERT), comments are labeled as Positive, Neutral, or Negative, and their distribution is computed per post.

Truth Level Categories:

- **Red:** >70% negative comments – high likelihood of being fake.
- **Orange:** 30–70% negative – mixed sentiment, uncertain credibility.
- **Green:** >70% positive – high likelihood of being true.



Figure 2. Truth Level Classification System

This sentiment-based classification provides indirect user validation and improves the model's ability to flag misinformation.

5.7 Model Training and Evaluation

5.7.1 Training Strategy

- The dataset is split **80/20 for training and testing** to assess generalization.
- **5-fold cross-validation** on LIAR and Facebook datasets yielded stable performance: **88.7% accuracy** ($\pm 1.1\%$), **87.9% precision** ($\pm 1.3\%$), and 86.8% F1-score ($\pm 1.2\%$).
- **Grid search** was used to tune learning rate, batch size, and neuron count.

5.8 Evaluation Metrics

To assess the performance of the proposed ANN model in fake news detection, several evaluation metrics were utilized. These metrics provide a comprehensive analysis of classification performance, ensuring both accuracy and robustness.

- **Accuracy**

Accuracy measures the overall effectiveness of the model in correctly classifying news articles as either fake or real. The ANN model achieved an accuracy of **89.2%**, significantly outperforming baseline models such as Logistic Regression (**76.4%**) and SVM (**81.3%**).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- **TP (True Positives)** = 4,315 (Fake news correctly classified)
- **TN (True Negatives)** = 3,980 (Real news correctly classified)
- **FP (False Positives)** = 630 (Real news misclassified as fake)
- **FN (False Negatives)** = 590 (Fake news misclassified as real)

– Precision and Recall

Precision and recall evaluate the model's ability to correctly identify fake news while minimizing false positives and false negatives.

- **Precision:** Measures how many of the predicted fake news articles were actually fake. The ANN model achieved a precision of **88.4%**, outperforming SVM (**80.6%**) and Logistic Regression (**75.2%**).

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{4,315}{4,315 + 630} = 88.4\%$$

- **Recall:** Measures the model's ability to correctly identify fake news without missing actual fake cases. The ANN model achieved a recall of **86.2%**, compared to SVM (**78.3%**) and Logistic Regression (**72.1%**).

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{4,315}{4,315 + 590} = 86.2\%$$

- **F1-Score**

The F1-score provides a balance between precision and recall, offering a single metric to evaluate the model's performance. The ANN model achieved an F1-score of **87.3%**, compared to SVM (**79.4%**) and Logistic Regression (**73.6%**).

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1 = 2 \times \frac{88.4 \times 86.2}{88.4 + 86.2} = 87.3\%$$

- **Confusion Matrix**

The confusion matrix provides insights into classification errors and model biases. Below is the confusion matrix for the ANN model:

Table 2. Existing Fake News Detection Methods

Actually / Predicted	Fake (Predicted)	Real (Predicted)
Fake (Actual)	4,315 (TP)	590 (FN)
Real (Actual)	630 (FP)	3,980 (TN)

- **AUC-ROC Score**

The **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)** score evaluates the model's ability to distinguish between fake and real news across different classification thresholds. The ANN model achieved an **AUC-ROC score of 0.91**, indicating a strong classification capability.

6. Results and Analysis

6.1 Performance of Deep Learning Models

The deep learning model tested in this study was evaluated on a combination of text-based features and sentiment-based feedback derived from user comments. The ANN model exhibited strong performance in terms of both accuracy and F1-score, outperforming traditional models such as logistic regression and support vector machines (SVM). The accuracy of the ANN model ranged between **85% to 90%**, with a precision rate of **88%** and recall of **86%**. These results indicate that the integration of user sentiment analysis provides a valuable source of additional information for fake news detection, improving the model's overall reliability.

6.2 Comparison with Baseline Models (Logistic Regression, SVM, and ANN.)

The comparison of the ANN model with baseline models, such as **Logistic Regression** and **Support Vector Machine (SVM)**, demonstrates the efficacy of deep learning approaches in detecting fake news. While the baseline models performed reasonably well, with **Logistic Regression** achieving an accuracy of **76%** and **SVM** reaching **81%**, the ANN model significantly outperformed them, as seen in the table below:

Table 3. Models Performance

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	76%	75%	72%	73%
Support Vector Machine (SVM)	81%	80%	78%	79%
Proposed ANN Model	89%	88%	86%	87%

6.3 Misinformation Propagation Patterns in Facebook Networks

The analysis of misinformation propagation across Facebook networks reveals distinct patterns of spread. Posts classified as fake news tend to show a higher degree of **engagement**, such as more comments, shares, and likes, compared to real news posts. Furthermore, fake news often exhibits a more rapid propagation rate, as users engage in debates or expres

6.4 Correlation Between Misinformation and User Engagement Metrics

The study further analyzed the correlation between user engagement metrics and misinformation. The key metrics included:

- **Number of Comments**
- **Sentiment Distribution of Comments**
- **Likes/Shares**

Fake news posts generally received a higher volume of comments, with a significant portion being **negative sentiments** (e.g., denials, rebuttals). These posts also experienced an increased number of shares, indicating that negative sentiment often drives higher engagement levels. Conversely, real news posts, although receiving positive feedback, saw lower engagement levels overall, as they are less likely to provoke heated discussions or emotional reactions from users.

7. Discussion

Combining content-based analysis with user sentiment significantly improves fake news detection accuracy. The color-coded truth-level system (Red, Orange, Green) offers a user-friendly way to interpret credibility, supporting real-time moderation. The study also confirms that emotionally charged and negative content drives faster misinformation spread, underscoring the need for platform-level interventions.

7.1 Effectiveness of AI-Based Detection

Deep learning models, particularly ANN, outperform traditional techniques by capturing complex linguistic and engagement patterns. Integrating sentiment analysis adds a valuable layer of context, making detection more nuanced and robust.

7.2 Challenges and Limitations

Despite promising results, challenges remain:

- **Bias:** Dataset imbalances may affect fairness; diverse data is needed.
- **Multilingual Issues:** Translation errors can reduce accuracy, especially for low-resource languages.
- **Real-Time Constraints:** Rapid misinformation spread demands faster, low-latency systems.

Although validated on three datasets, future testing across platforms (e.g., Twitter, Reddit) is needed for broader generalization.

8. Ethical Considerations

Using public social media data requires strong ethical safeguards. Personal data must not be used without consent, and transparency is essential to maintain trust. AI can significantly aid fake news detection, but ethical

deployment must address bias, privacy, and fairness. Collaboration among researchers, platforms, and policymakers is crucial for long-term, responsible impact.

9. Future Directions

Future research will focus on enhancing **multilingual sentiment detection** by leveraging advanced models like mBERT and XLM-R to support diverse languages. Efforts will also target **real-time detection** through the development of fast approximate nearest neighbor (ANN) models integrated with live fact-checking APIs. **Behavior-aware models** using graph neural networks (GNNs) and reinforcement learning will be applied to monitor and adapt to evolving misinformation patterns. Improving **sentiment accuracy**, especially in recognizing sarcasm and emotional context, will be prioritized, alongside the exploration of dashboard-based visualizations to aid interpretation. **Privacy-preserving methods** such as federated learning and differential privacy will be employed to ensure secure and ethical AI deployment. Finally, **policy integration** through collaboration with social media platforms and regulators will be essential to embed these AI tools effectively into content moderation workflows.

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